

Chapter 22

Lecture 22: The Tychonoff theorem.

Idea For the idea of this lecture, refer to Munkres' Topology, P230 or pdf P247.

Definition 22.0.1. Let $(A, \leq)$ be a partially ordered set. A subset $B \subseteq A$ is called a chain if for any $a, b \in B$, we have either $a \leq b$ or $b \leq a$.

Theorem 22.0.1 (Zorn's lemma). Let $(A, \leq)$ be a partially ordered set. If every chain in A has an upper bound in A , then A contains at least one maximal element.

Lemma 22.0.1. Let X be a set; let $\mathcal{A}$ be a collection of subsets of X having the finite intersection property. Then there is a collection $\mathcal{D}$ of subsets of X such that $\mathcal{D}$ contains $\mathcal{A}$, and $\mathcal{D}$ has the finite intersection property, and no collections of subsets of X properly containing $\mathcal{D}$ has the finite intersection property.

Proof

For purposes of this proof, we shall call a set whose elements are collections of subsets of X a "superset" and shall denote it by an outline letter. To summarize the notation:

- c is an element of X ;
- C is a subset of X ;
- $\mathcal{C}$ is a collection of subsets of X ;
- $\mathbb{C}$ is a superset whose elements are collections of subsets of X .

Now by hypothesis, $\mathcal{A}$ is a collection of subsets of X having the finite intersection property. Let $\mathbb{A}$ be the superset consisting of all collections of subsets of X such that $\forall \mathcal{B} \in \mathbb{A}$, we have $\mathcal{B} \supseteq \mathcal{A}$ and $\mathcal{B}$ has the finite intersection property. To prove the lemma, we need to show that $\mathbb{A}$ contains a maximal element with respect to the partial order $\subseteq$.

In order to apply Zorn's lemma, we must show that if $\mathbb{B}$ is a "subsuperset" of $\mathbb{A}$ that is a chain, then $\mathbb{B}$ has an upper bound in $\mathbb{A}$. Let

$$\mathcal{C} = \bigcup_{\mathcal{B} \in \mathbb{B}} \mathcal{B}. \quad (22.1)$$

Certainly $\mathcal{C}$ contains $\mathcal{A}$, since each $\mathcal{B} \in \mathbb{B}$ contains $\mathcal{A}$. Let $C_1, C_2, \dots, C_n$ be finitely many elements of $\mathcal{C}$. Then for each $i = 1, 2, \dots, n$, there exists $\mathcal{B}_i \in \mathbb{B}$ such that $C_i \in \mathcal{B}_i$. Since $\mathbb{B}$ is a chain, there exists $\mathcal{B}_j \in \mathbb{B}$ such that $\mathcal{B}_i \subseteq \mathcal{B}_j$ for all $i = 1, 2, \dots, n$. Thus $C_i \in \mathcal{B}_j$ for all $i = 1, 2, \dots, n$. Since $\mathcal{B}_j$ has the finite intersection property, we have

$$C_1 \cap C_2 \cap \dots \cap C_n \neq \emptyset. \quad (22.2)$$

Thus $\mathcal{C}$ has the finite intersection property. And so $\mathcal{C} \in \mathbb{A}$. Moreover, for each $\mathcal{B} \in \mathbb{B}$, we have $\mathcal{B} \subseteq \mathcal{C}$. Thus $\mathcal{C}$ is an upper bound of $\mathbb{B}$ in $\mathbb{A}$. By Zorn's lemma, $\mathbb{A}$ contains a maximal element, which we denote by $\mathcal{D}$. This completes the proof of the lemma. ■

Lemma 22.0.2. Let X be a set; let $\mathcal{D}$ be a collection of subsets of X that is maximal with respect to the finite intersection property. Then

1. Any finite intersection of elements of $\mathcal{D}$ is an element of $\mathcal{D}$;
2. If A is a subset of X that intersects every element of $\mathcal{D}$, then A is an element of $\mathcal{D}$.

Proof to 1

Let B be any finite intersection of elements of $\mathcal{D}$. Let $\mathcal{E} = \mathcal{D} \cup \{B\}$. We can show that $\mathcal{E}$ has the finite intersection property (Check it). By maximality of $\mathcal{D}$, we have $\mathcal{E} = \mathcal{D}$, so $B \in \mathcal{D}$.

Proof to 2

Let A be a subset of X that intersects every element of $\mathcal{D}$. Let $\mathcal{E} = \mathcal{D} \cup \{A\}$. Take finitely many elements from $\mathcal{E}$. If none of them is A , then their intersection is not empty because $\mathcal{D}$ has the finite intersection property, otherwise, it is of the form

$$A \cap D_1 \cap D_2 \cap \dots \cap D_n, \quad (22.3)$$

Now, $D_1 \cap D_2 \cap \dots \cap D_n \in \mathcal{D}$ by part 1, so A intersects it by hypothesis. Thus the intersection is not empty. So $\mathcal{E}$ has the finite intersection property. By maximality of $\mathcal{D}$, we have $\mathcal{E} = \mathcal{D}$, so $A \in \mathcal{D}$. ■

Theorem 22.0.2 (Tychonoff theorem). *Any arbitrary product of compact spaces is compact in the product topology.*

Proof

Let

$$X = \prod_{\alpha \in J} X_{\alpha}, \quad (22.4)$$

where each X_{α} is compact. Let $\mathcal{A}$ be a collection of subsets of X having the finite intersection property. We prove that the intersection $\bigcap_{A \in \mathcal{A}} A$ is non-empty. Compactness of X follows.

By 22.0.1, we choose $\mathcal{D}$ containing $\mathcal{A}$ that is maximal with respect to the finite intersection property. It suffices to show that the intersection

$$\bigcap_{D \in \mathcal{D}} \bar{D} \quad (22.5)$$

is non-empty.

Given any $\alpha \in J$, we consider the collection

$$\{\pi_{\alpha}(D) \mid D \in \mathcal{D}\} \quad (22.6)$$

of subsets of X_{α} . This collection has the finite intersection property because $\mathcal{D}$ does. By compactness of X_{α} , we can choose a point $x_{\alpha} \in X_{\alpha}$ such that

$$x_{\alpha} \in \bigcap_{D \in \mathcal{D}} \overline{\pi_{\alpha}(D)}. \quad (22.7)$$

Now the point $x = (x_{\alpha})$ is defined in X .

Let U_{β} be a neighborhood of x_{β} in X_{β} . Since $x_{\beta} \in \overline{\pi_{\beta}(D)}$, we have that $\exists y \in D, \pi_{\beta}(y) \in U_{\beta} \cap \pi_{\beta}(D)$. Thus $y \in \pi_{\beta}^{-1}(U_{\beta}) \cap D$. Since this is true for any $D \in \mathcal{D}$, by the previous lemma, we have $\pi_{\beta}^{-1}(U_{\beta}) \in \mathcal{D}$. So every subbasis element containing x is in $\mathcal{D}$. And then it follows from the same lemma that every basis (finite intersection of subbasis) element containing x is in $\mathcal{D}$. Thus any open neighborhood of x intersects every $D \in \mathcal{D}$. Therefore, $x \in \bar{D}$ for all $D \in \mathcal{D}$. ■